Worked Answer Key

For inferred patterns and table rules, accept other clearly explained rules that fit every given value and any stated conditions, with matching answers.

1. **P1:** 1) Add 5: 27, 32. 2) Subtract 5: 25, 20. 3) Add 10: 146, 156.
2. **P2:** 1) Add 7: 31, 38. 2) Subtract 8: 60, 52. 3) Add 4: 5, 9, 13, 17, 21 chairs.
3. **P3:** 1) Multiply by 3: 162, 486. 2) Divide by 2: 20, 10. 3) Multiply by 4: 256, 1024.
4. **P4:** 1) $3(7) + 2 = 23$. 2) Multiply the input by 3, then add 2: $3(6) + 2 = 20$. 3) Multiply by 3: $3(10) = 30$.
5. **P5:** 1) $y = 9, 13, 20$. 2) $y = 2, 6, 14$, so $(1, 2), (3, 6), (7, 14)$. 3) Each increase of 3 in x gives an increase of 9 in y. A matching rule is $y = 3x + 1$, so $y = 3(11) + 1 = 34$.
6. **P6:** 1) Multiply x by 2, then add 4: $y = 2x + 4$. 2) Divide x by 2, then multiply by 5: $y = (x \div 2) \times 5$ (equivalently, $y = 2.5x$). 3) Multiply x by 5, then add 12: $y = 5x + 12$.
7. **P7:** 1) $9 + 14 = 23$. 2) $4(7) = 28$. 3) $3(6) + 2 = 20$.
8. **P8:** 1) $q + 8$. 2) $5r$. 3) $t - 6$.
9. **P9:** 1) Let b be the number of pencils per box: $7b$ pencils. 2) Let d be Maya's starting amount in dollars: $d + 12$ dollars. 3) Let p be the number of pages read: $40 - p$ pages left.
10. **P10:** 1) $x = 43 - 18 = 25$. Check: $25 + 18 = 43$. 2) $a = 60 - 27 = 33$. Check: $33 + 27 = 60$. 3) $y = 52 - 15 = 37$. Check: $15 + 37 = 52$.
11. **P11:** 1) $z = 29 + 16 = 45$. Check: $45 - 16 = 29$. 2) $c = 75 - 38 = 37$. Check: $75 - 37 = 38$. 3) $h = 19 + 24 = 43$. Check: $43 - 24 = 19$.
12. **P12:** 1) $x = 45 \div 5 = 9$. 2) $n = 63 \div 7 = 9$. 3) $p = 28 \div 4 = 7$.
13. **P13:** 1) $q = 8(6) = 48$. 2) $r = 72 \div 9 = 8$. 3) $s = 11(5) = 55$.
14. **P14:** 1) $n + 35 = 82$, so $n = 82 - 35 = 47$. The number is 47. 2) $6a = 54$, so $a = 54 \div 6 = 9$ apples per bag. 3) $s - 18 = 27$, so $s = 27 + 18 = 45$ stickers to start.
15. **P15:** 1) Terms: 6, 15, 24, 33, 42, 51; term 6 is 51. 2) 3, 6, 12, 24, 48, 96; terms 5 and 6 are 48 and 96. 3) Terms: 30, 27, 24, 21, 18, 15, 12, 9; term 8 is 9.
16. **P16:** 1) Add 7; term 5 is 39. 2) Multiply by 3; next is 108. 3) Missing 35; subtract 5.
17. **P17:** 1) A = 18, B = 16; A is greater. 2) A = 15, B = 17; B is greater. 3) B is always greater by 4 because both share $4x$.
18. **P18:** 1) Rule $8 + 2(n - 1)$; row 7 has 20 seats. 2) $14 + 3(6) = 32$ cm. 3) $4c = 20$; $c = 5$ dollars.
19. **P19:** 1) Multiply by 3: 135, 405. 2) $2(9) + 7 = 25$. 3) $w = 71 - 26 = 45$.

20. **P20:** 1) Term n is $9+6(n-1)$ because there are $n-1$ additions after term 1. Term 10 is $9+6(9)=63$. 2) $y=4,10,19$. 3) Let m be Mia's number of cards. Then $2m+5=23$. Subtract 5 to get 18, then divide by 2: $m=9$ cards.